

Investigation of helium injection cooling to liquid oxygen under pressurized condition

Namkyung Cho ^{a,*}, Ohsung Kwon ^a, Youngmog Kim ^a, Sangkwon Jeong ^{b,1}

^a *Propulsion Control Department, Korea Aerospace Research Institute, 45, Oun-Dong, Yuseong-gu, Daejeon 305-333, Republic of Korea*

^b *Cryogenic Engineering Laboratory, Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology, Guseong-dong, Yuseong-gu, Daejeon 305-701, Republic of Korea*

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Abstract

Sub-cooling of cryogenic propellant by helium injection is one of the most effective methods for suppressing bulk boiling and keeping sub-cooled liquid oxygen before rocket launch. Compared with the helium injection cooling under atmospheric condition, helium injection cooling under pressurized condition has advantage that it can greatly reduce re-warming time of the sub-cooled liquid oxygen. Helium injection cooling under pressurized condition is characterized by cooling of initially sub-cooled cryogenic liquid, which is significantly different from that of the atmospheric condition where liquid oxygen usually exists at saturated condition. In this paper, we discuss the characteristics of helium injection cooling under pressurized condition, with the associated physical understanding of the process. Experimental results are presented along the simulations of variously combined system parameters based on the finite heat transfer and instantaneous diffusion mass transfer model. A non-dimensional parameter for identifying the cooling regime is conceived. The critical values of the non-dimensional parameters and injected helium temperatures are also estimated.

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1. Introduction

Sub-cooling of cryogenic propellant by helium injection is known to be one of the most effective methods for suppressing bulk boiling and keeping liquid oxygen sub-cooled before rocket launch. By the helium gas injected through liquid oxygen, large surface area for evaporation and diffusion of oxygen into bubbles can be realized. A similar concept is also applied for ground pre-pressurization for upper stages [1,2]. Over the past few decades, several studies have been done on this subject. Larsen and Clark have shown the possible maximum sub-cooling by assuming the injected helium and the oxygen vapor to be in phase equilib-

rium [3]. Schmidt has achieved sub-cooling of liquid hydrogen by helium injection [4]. Cleary et al. have sub-cooled liquid oxygen by injected helium with various injection flow rates as a part of liquid oxygen conditioning research [5]. Recently, Cho et al. have applied finite rate heat transfer and instantaneous mass transfer model considering the limitation of heat transfer in a bubbling system [6].

The previous studies, however, were all based on tests and analysis that were performed under atmospheric conditions. For practical application, it is helpful to understand the characteristics of helium injection cooling under pressurized condition which is observable in pre-pressurization period of rocket. Liquid rocket launch scenario usually includes several stages such as chilling tank, filling cryogenic propellant, holding, pre-pressurization, and firing. Helium injection to cryogenic propellant has been usually applied prior to pre-pressurization. For that launch scenario, the sub-cooled cryogenic propellant may be

* Corresponding author. Tel.: +82 42 860 2336; fax: +82 42 860 2698.

E-mail addresses: cho@kari.re.kr (N. Cho), oskwon@kari.re.kr (O. Kwon), ymkim@kari.re.kr (Y. Kim), skjeong@kaist.ac.kr (S. Jeong).

¹ Fax: +82 42 869 3210.

A_s	net bubble surface area (m ²)	$\dot{Q}_{amb}$	heat transfer rate to liquid oxygen through tank wall from ambient and ullage region (W)
$A_{b,d}$	detached bubble surface area in spherical form (m ²)	$\dot{Q}_{gl}$	heat transfer rate from gas in bubble to liquid oxygen (W)
$C_{p,LOX}$	heat capacity of liquid oxygen (kJ/kg K)	R	universal gas constant (8.314 kJ/kmol K)
D	diameter (mm)	Ra	Rayleigh number
$\overline{D}_b$	average bubble diameter (mm)	r_b	bubble radius (mm)
$D_{b,d}$	detached bubble diameter in spherical form (mm)	t	time (s)
D_{O_2-He}	diffusion coefficient for oxygen flowing through helium (m ² /s)	T	temperature (K)
f_b	bubble formation frequency (1/s)	T_g	gas temperature (K)
f'_b	number of bubbles generated during initial 1 s ($f_b \cdot 1$ s)	$T_{He,crit}$	critical helium temperature defined in Eq. (31) (K)
h_{gi}	gas-to-interface heat transfer coefficient (W/m ² K)	T_{ref}	normal boiling-point temperature at P_{ref} (K)
h_{li}	liquid-to-interface heat transfer coefficient (W/m ² K)	$T_{sat}(P_{GOX,b})$	saturation temperature of the liquid oxygen at $P_{GOX,b}$ (K)
$\overline{h}_{gi}$	average gas-to-interface heat transfer coefficient (W/m ² K)	T_i	interface temperature (K)
$\overline{h}_{li}$	average liquid-to-interface heat transfer coefficient (W/m ² K)	T_{LOX}	liquid oxygen temperature (K)
h_{fg}	heat of vaporization of liquid oxygen (kJ/kg)	$T_{LOX,i}$	initial liquid oxygen temperature (K)
H_{fg}	heat of vaporization of liquid oxygen per mole (kJ/mole)	T_w	wall temperature (K)
i	real interface	u_b	bubble rising velocity (m/s)
m_{LOX}	mass of liquid oxygen (kg)	v	diffusion volume in Eq. (6)
$\dot{m}_{He}$	helium injection mass flow rate (kg/s)	$V_{b,d}$	detached bubble volume in Eq. (8) (m ³)
$\dot{m}_{LOX,h}$	evaporation rate of liquid oxygen due to heat flux (kg/s)	V_{LOX}	LOX volume (m ³)
$\dot{m}_{LOX,diff}$	evaporation rate of liquid oxygen due to diffusion (kg/s)	$\dot{V}_{He}$	helium injection volume flow rate (m ³ /s)
$\dot{m}_{GOX}$	total generation rate of oxygen vapor ($\dot{m}_{GOX} = \dot{m}_{LOX}$) (kg/s)	x_{O_2}	oxygen mole fraction
$\dot{m}_{LOX}$	total evaporation rate of liquid oxygen ($\dot{m}_{LOX,h} + \dot{m}_{LOX,diff}$) (kg/s)	$x_{O_2,u}$	oxygen mole fraction at ‘u’ surface
M	molecular weight (kg/kmol)	$x_{O_2,s}$	oxygen mole fraction at ‘s’ surface
M_{GOX}	molecular weight of oxygen (kg/kmol)	X	mole ratio defined in Eq. (16)
M_{He}	molecular weight of helium (kg/kmol)	X_{cr}	critical value of X for cooling of liquid oxygen defined in Eq. (30)
n	mole	Z	compressibility factor
$\dot{n}$	mole change rate (1/s)		
$n_{He,b}$	moles of helium gas in bubbles		
$n_{GOX,b}$	moles of oxygen vapor in bubbles		
$\overline{Nu}_D$	average Nusselt number		
P	pressure (MPa)		
P_{GHe}	partial pressure of helium gas (MPa)		
$P_{GHe,b}$	partial pressure of helium gas in bubbles (MPa)		
P_{GOX}	partial pressure of oxygen gas (MPa)		
$P_{GOX,b}$	partial pressure of oxygen gas in bubbles (MPa)		
$P_{GOX,s}$	partial pressure of oxygen gas at ‘s’ surface		
P_{ref}	reference pressure (atmospheric pressure) (MPa)		
$P_{sat}(T_{LOX})$	saturation vapor pressure of liquid oxygen at T_{LOX} (MPa)		
P_{sys}	system pressure (MPa)		

<i>Greek letters</i>	
α	void fraction, thermal diffusivity (m ² /s)
β	coefficient of thermal expansion (1/K)
μ	dynamic viscosity (Pa s)
ν	kinematic viscosity of (m ² /s)
ρ_{He}	helium density (kg/m ³)
$\sum$	summation symbol
σ	surface tension of liquid (N/m)

<i>Subscripts</i>	
amb	ambient
b	bubble
g	gas
gi	gas-to-interface
GOX	gaseous oxygen
He, GHe	helium gas
i	real interface, initial value
L	liquid
li	liquid-to-interface
LOX	liquid oxygen
s	vapor side interface

sat	saturation
sys	system
u	liquid side interface
w	wall

Abbreviations

GHe	gaseous helium
GOX	gaseous oxygen

LN2	liquid nitrogen
LOX	liquid oxygen
min	minute
TT	temperature transmitter
PT	pressure transmitter
LT	level transmitter
FM	flow meter

re-warmed by the heat transfer during pre-pressurization stage. If the helium injection cooling is performed after pre-pressurization, the temperature of cryogenic propellant can be maintained at low level.

For the purpose of ground pre-pressurization with helium bubbling, it is desirable for liquid oxygen bulk temperature to decrease or at least not increase during this process. To satisfy the required liquid oxygen temperature, proper combination of relevant operating parameters should be identified.

In this paper, we discuss the characteristics of helium injection cooling under pressurized condition, with the associated physical understanding of the process. Experimental results along the simulations of variously combined system parameters are presented. A non-dimensional parameter for identifying the cooling regime is also introduced.

This paper is organized as follows. Section 2, starting with the principle of helium injection cooling, reviews the finite heat transfer and instantaneous diffusional mass transfer concept developed in the authors' previous article [6]. In Section 3, we discuss the qualitative characteristics of helium injection cooling under pressurized condition. Section 4 describes the method of analysis and the experiment. Section 5 presents the results and discussion of the analysis and experiments. Finally, in Section 6, a non-dimensional parameter for identifying cooling regime is conceived. The critical values of the non-dimensional parameters and the injected helium temperatures are also estimated.

2. Principle of helium injection cooling and its modeling

Fig. 1 shows the schematic of helium injection cooling system. Fundamental principle of sub-cooling of cryogenic propellant by helium injection is diffusional mass transfer. Since the injected helium is pure, the difference in the partial pressure of oxygen in helium bubble and the vapor pressure of liquid oxygen causes diffusional mass transfer of oxygen vapor into the helium bubble. Ideally, the oxygen continues to diffuse into the helium bubbles until thermodynamic phase equilibrium is reached. The equilibrium point is established when the partial pressure of oxygen vapor in the bubble $P_{GOX,b}$ is exactly equal to the

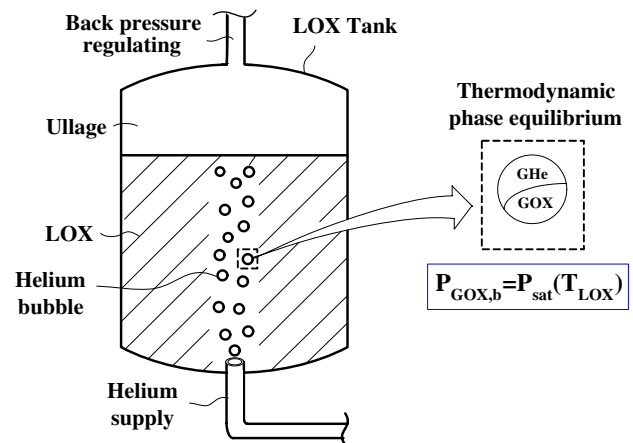


Fig. 1. Schematic diagram of helium injection cooling system.

saturated vapor pressure of liquid oxygen at particular temperature of the liquid $P_{sat}(T_{LOX})$. This diffusion process (or evaporation process) results in a cooling effect caused by the vaporization heat of surrounding liquid.

Fig. 2 portrays the sub-cooling effect of helium injection. In order to clearly define the partial pressure at the interface surface, imaginary surfaces denoted by 'u' and 's' are introduced on both sides of the real interface. Each surface is indefinitely close to the real interface i , as shown in

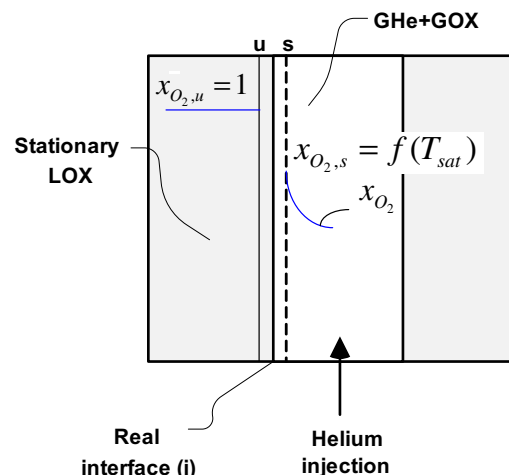


Fig. 2. Schematic diagram of principle on helium injection cooling.

Fig. 2, with liquid oxygen evaporating into helium gas stream. At ‘u’ surface, liquid-phase quantities of oxygen are assumed and mole fraction of oxygen $x_{O_2,u}$, while, at ‘s’ surface, gas phase quantities of oxygen are assumed and mole fraction $x_{O_2,s}$ is calculated from thermodynamic equilibrium with the liquid oxygen at the ‘u’ surface. If pure helium is injected into liquid oxygen, the partial pressure of oxygen vapor at the ‘s’ surface would significantly differ from that in bulk gas, which causes diffusional mass transfer of oxygen according to Fick’s law of diffusion from the ‘s’ surface. In this case liquid oxygen should be vaporized and supplied to the gas region to satisfy thermodynamic phase equilibrium in the gas region. The total liquid oxygen vaporization rate $\dot{m}_{LOX}$ may be expressed by the summation of two types of vaporization according to their origins: the vaporization rate due to heat transfer between the gas bubble and liquid oxygen $\dot{m}_{LOX,h}$ and the vaporization rate due to pure diffusion $\dot{m}_{LOX,diff}$ as shown in Eq. (1). If the amount of vaporization is small compared with the remaining liquid oxygen mass, $\dot{m}_{LOX} \cdot \Delta t / m_{LOX} \ll 1$, then the rate at which the bulk liquid oxygen energy change is given as in Eq. (2), where $\dot{Q}_{gi}$ is the heat transfer rate from the gas bubble to the liquid oxygen interface, $\dot{m}_{LOX} h_{fg}$ is the latent heat transfer rate by phase change of liquid oxygen, and $\dot{Q}_{amb}$ is the heat transfer rate into liquid oxygen from surrounding ambient. The liquid oxygen vaporization due to heat transfer $\dot{m}_{LOX,h}$ can be related to $\dot{Q}_{gi}$ as shown in Eq. (3). Substituting Eqs. (1) and (3) into the energy balance equation, Eq. (2), the vaporization due to heat transfer $\dot{m}_{LOX,h}$ is now canceled. Therefore, the vaporization due to diffusion $\dot{m}_{LOX,diff}$ is the main drive for cooling. Liquid oxygen sub-cooling by helium injection can be conclusively characterized by diffusion-driven evaporative cooling

$$\dot{m}_{LOX} = \dot{m}_{LOX,h} + \dot{m}_{LOX,diff}, \quad (1)$$

$$m_{LOX} C_{p,LOX} \frac{dT_{LOX}}{dt} = \dot{Q}_{gi} - \dot{m}_{LOX} h_{fg} + \dot{Q}_{amb}, \quad (2)$$

$$\dot{Q}_{gi} \approx \dot{m}_{LOX,h} h_{fg}. \quad (3)$$

In the investigation of helium injection cooling, three limiting cases can be established in terms of heat and mass transfer aspects. The first case is instantaneous heat and mass transfer where the bulk liquid and the gas are always in thermodynamic phase equilibrium state. This approach theoretically gives maximum liquid oxygen evaporation rate and sub-cooling rate. The second case is finite heat transfer and instantaneous diffusional mass transfer. In this case, the temperatures of the bulk liquid and gas are different due to finite heat transfer. However, due to the assumption of infinite diffusion mass transfer rate, the partial pressure of the oxygen vapor at the ‘s’ surface remains equal to that of the bulk gas. The third case is finite heat and mass transfer. Although the ‘finite heat and mass transfer model’ is certainly the most realistic description of the phenomenon, the ‘finite heat transfer

and instantaneous diffusional mass transfer model’ gives satisfactory agreement with experimental data for small bubbles and gas with high diffusion coefficient such as helium [6].

For the finite heat transfer and instantaneous diffusional mass transfer model, liquid oxygen evaporation rate $\dot{m}_{LOX}$ is calculated with the prescribed heat transfer coefficients (gas-to-interface: h_{gi} , liquid-to-interface: h_{li}), net bubble surface area (A_s) and interface temperature (T_i) for a given system as shown in Eq. (4) [6]

$$\dot{m}_{LOX} = A_s (h_{gi}(T_g - T_i) - h_{li}(T_i - T_{LOX})) / h_{fg}. \quad (4)$$

In Eq. (4), with the instantaneous diffusion mass transfer assumption, the partial pressure of oxygen at the ‘s’ surface, $P_{GOX,s}$ is equal to the bulk partial pressure of oxygen in the bubble, $P_{GOX,b}$. Also, since the ‘s’ surface is indefinitely close to the real interface (i), the real interface temperature T_i in Eq. (4) can be calculated from the partial pressure of oxygen in the bubble as in Eq. (5)

$$T_i = T_{sat}(P_{GOX,b}). \quad (5)$$

Liquid oxygen has finite vaporization rate depending on the net bubble surface area, and the relevant heat transfer coefficients. Since thermodynamic phase equilibrium in bubble requires the partial pressure of oxygen in the bubble to be equal to the saturated vapor pressure of liquid oxygen at particular temperature of liquid, the difference between the partial pressure of the oxygen in the bubble and the vapor pressure of the liquid oxygen propels the generated oxygen vapor into the bubble without degradation (not diffusion-limited). The transferred oxygen vapor mixes with helium gas instantaneously.

3. Characteristics of helium injection cooling under pressurized condition

To understand the characteristics of helium injection cooling under pressurized condition, a comparison with the case of atmospheric condition may be helpful.

First, liquid oxygen is initially at sub-cooled state under pressurized condition. This is significant difference from the atmospheric condition where liquid oxygen usually exists at saturated condition (90.2 K at 1 bar). To provide proper propellant temperature to engine, cryogenic propellant should maintain its sub-cooled state during pressurized period. Therefore, helium injection cooling under pressurized condition must be characterized by cooling of initially sub-cooled cryogenic liquid.

The main difference between saturated and sub-cooled liquid is the vapor pressure. For saturated condition, the liquid oxygen vapor pressure is equal to its system pressure ($P_{sat}(T_{LOX})/P_{sys} = 1$), whereas for sub-cooled condition, the liquid oxygen vapor pressure is lower than the system pressure ($P_{sat}(T_{LOX})/P_{sys} < 1$). Since the pressure difference

$P_{\text{sat}}(T_{\text{LOX}}) - P_{\text{GOX,b}}$ causes diffusional mass transfer, the lower liquid oxygen vapor pressure reduces the effectiveness of helium injection cooling. This feature is deeply discussed in later sections with analytical and experimental results.

Second, since the system pressure is high, relatively higher helium mass flow rate is supplied with a specified volume flow rate. In general, the volume flow rate of helium $\dot{V}_{\text{He}}$ in the case of helium injection cooling is limited by the situation for individual bubbles not to have interaction which significantly decrease the overall bubble surface area by conglomeration. Therefore, $\dot{V}_{\text{He}}/V_{\text{LOX}}$ usually has a limited range of variation among systems under different system pressures associated with helium injection cooling. For the specified volume flow rate, the helium mass flow rate is $\dot{m}_{\text{He}} = \rho_{\text{He}} \dot{V}_{\text{He}}$, where the helium mass flow rate $\dot{m}_{\text{He}}$ increases with density ρ_{He} which also increases with the system pressure under pressurized condition. The increased helium mass flow rate lowers the partial pressure of oxygen in the helium bubble, which enhances diffusional mass transfer. However, as the mass flow rate increases, so does the amount of energy flow, which increases liquid oxygen evaporation due to heat transfer and always negatively contributes to the desired cooling. The effects of increased liquid oxygen evaporation due to increased energy flow are extensively discussed in the following sections.

Third, the diffusion coefficient is known to be inversely proportional to the system pressure as described in Eq. (6) [7],

$$D_{\text{GOX-GHe}} = \frac{10^{-3} \cdot T^{1.75} \cdot [(M_{\text{GOX}} + M_{\text{GHe}})/(M_{\text{GOX}}M_{\text{GHe}})]^2}{P \cdot [(\sum v)_{\text{GOX}}^{1/3} + (\sum v)_{\text{GHe}}^{1/3}]^2}. \quad (6)$$

We will not consider the pressure effect on the diffusion coefficient in this paper.

4. Methods of analysis and experiment

4.1. Analysis

The analysis procedure consists of two steps: the single bubble analysis and the bubbling system analysis. From the single bubble analysis, the average bubble diameter $\bar{D}_b$, heat transfer coefficients (gas-to-interface heat transfer coefficient; $\bar{h}_{\text{gi}}$ and liquid-to-interface heat transfer coefficient; $\bar{h}_{\text{li}}$) and average bubble rising velocity $\bar{u}_b$ of single bubble rising through liquid oxygen during bubble residence time in liquid oxygen are obtained for the specified helium injection cooling system. The detailed description of correlations for each step of single bubble analysis is available in Ref. [6].

The parameter values obtained from the single bubble analysis are used for the bubbling system analysis. Table 1 shows two input conditions for the bubbling system analysis under pressurized condition (3 bar): sub-cooled (91 K) and saturated (102 K), which are obtained from the single bubble analysis.

The bubbling system analysis is performed using a commercial code SINDA/FLUINT, the comprehensive finite-difference, lumped parameter tool [8]. To model non-equilibrium phase bubbler with the concept of finite heat transfer and instantaneous diffusion, two lumps are established: one liquid lump and one vapor lump.

The pressure relation is applied for the pair of two lumps as shown in Eq. (7) considering the gas lump as a spherical bubble

$$P_g = P_l + \frac{2\sigma}{r_b}. \quad (7)$$

The distribution of bubble size or rising velocity is not considered in this bubbling system. We use simply a single

Table 1
Examples of input parameters for bubbling system analysis

Parameters	Sub-cooled condition	Saturated condition	Ref.
System pressure	3 bar	3 bar	P_{sys} is maintained by atmospheric venting
Initial LOX temperature	91 K	102 K	
Helium injection rate ($\dot{V}_{\text{He}}/V_{\text{LOX}}$)	0.0012 m ³ /s (1.2 min ⁻¹)	0.0014 m ³ /s (1.4 min ⁻¹)	
Injected Helium Temperature	288 K	288 K	
Detached bubble diameter ($D_{b,i}$)	0.056 m	0.059 m	
Average bubble diameter ($\bar{D}_b$)	0.051 m	0.052 m	
Average rising bubble velocity (m/s)	0.54 m/s	0.55 m/s	
Average bubble volume ($\bar{V}_b$)	0.071	0.0741	$\bar{V}_b = 4/3 \times (\bar{D}_b/2)^3$
Number of bubble (N_b)	$N_b = V_g/\bar{V}_b$		Total gas volume (V_g) depends on helium injection rate and LOX vaporization
Overall bubble surface area (A_s)	$N_b \times 4 \times \pi \times (\bar{D}_b/2)^2$		
Average gas-to-interface heat transfer coefficient ($\bar{h}_{\text{gi}}$)	19.8 W/m ² K	37.8 W/m ² K	
Average liquid-to-interface heat transfer coefficient ($\bar{h}_{\text{li}}$)	443.2 W/m ² K	398.1 W/m ² K	

diameter and a single rising velocity for numerical analysis under the assumption that bubbles can be treated independently from other bubbles, which is valid for low void fraction system ($\alpha < 0.1$) [9].

The interface area corresponding to overall bubble surface area is calculated based on the number of bubbles in the bubbling system. The number of bubbles is obtained from the ratio of the mixture gas volume composed of helium and oxygen vapor to the averaged single bubble volume obtained from the single bubble analysis. The number of bubbles multiplied by the averaged single bubble surface area gives the overall surface area at specific time. The average heat transfer coefficients obtained from the single bubble are used for the bubbling system analysis. With given thermodynamic states (temperatures, pressures, partial pressures, etc.), the overall surface area and the heat transfer coefficients of single bubble, the instantaneous oxygen vaporization rate is readily calculated with Eq. (4). The interface temperature is calculated based on the partial pressure of oxygen in the gas region. The generated oxygen vapor in liquid lump is assumed to be propelled into the gas lump immediately and mixed with helium gas. During the helium injection process, the gas lump pressure is kept constant by adjusting loss factor of vent valve in SINDA/FLU-INT code. This simulation algorithm calculates the cooling performance and the related data for helium injection cooling of liquid oxygen with the assumption of finite heat transfer and instantaneous diffusion mass transfer.

4.2. Test methods

The injection cooling process has been investigated experimentally using two geometrically and functionally different test setups. The functional difference between the two test facilities is the injected helium temperature: one at ambient temperature and the other at low temperature.

Fig. 3 shows the experimental setup for a 90 l, 2.5 m tall by 0.23 m diameter test tank into which ambient temperature helium is injected. Axial temperature distribution is measured with 10 RTD type temperature sensors. Fig. 4 shows the experimental setup for a 60 l test tank with a 0.2 m inside diameter and 2.0 m height. Helium gas is cooled by a heat exchanger immersed in liquid nitrogen bath. The axial temperature distribution is measured with 18 RTD type temperature sensors placed at 10 cm interval along the height. For the two experimental facilities, average temperature transients are plotted in the following results, the deviation from the average value along axial direction of the tank is within 0.2 K. It is considered that the thermal stratification between layers of liquid oxygen is small due to the mixing effect caused by the injected helium gas.

For the two test setups, the tank was filled with liquid oxygen up to 80% of the tank height prior to helium injection. Helium gas at constant volume flow rate was injected through a submerged orifice with 6.3 mm diameter by adjusting the upstream pressure referred by the measured

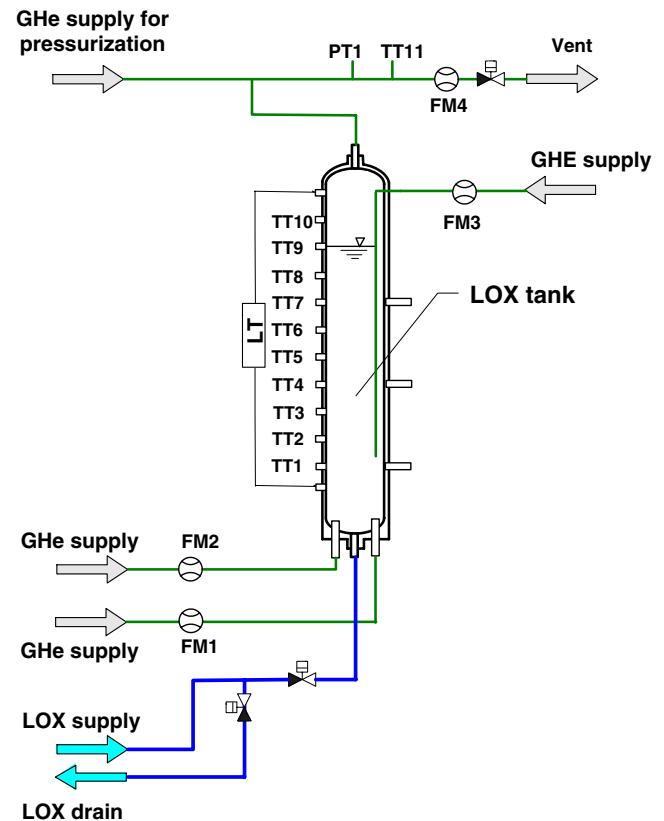


Fig. 3. Schematics of test setup for ambient temperature helium supply.

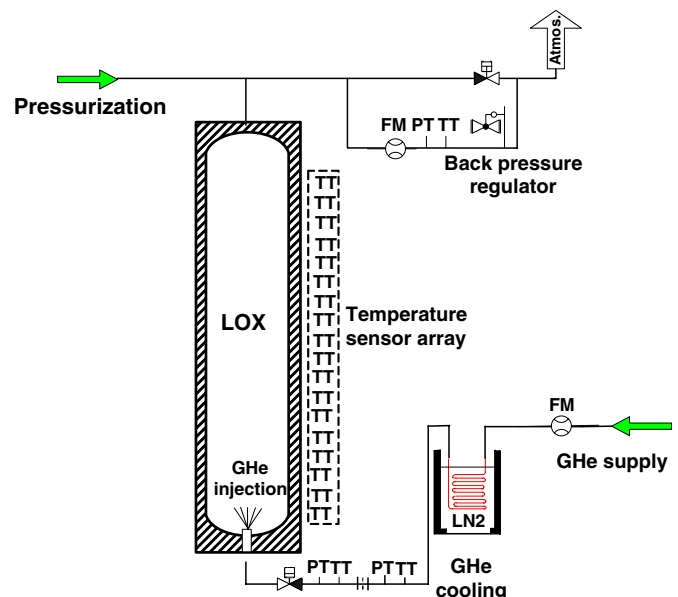


Fig. 4. Schematics of test setup with low temperature helium supply.

volume flow rate value of a turbine type flow-meter. The tank pressure was controlled by a back pressure regulator installed at the vent port and the ullage pressure increment was measured. Before each helium injection test, ambient background heating loss was calculated from the measured

Table 2
Main parameters and their effects for helium injection cooling

No.	Variable parameters	Affected parameters
1	Ratio of system (LOX) pressure to LOX saturation pressure temperature corresponding to LOX temperature, $P_{\text{sys}}/P_{\text{sat}}(T_{\text{LOX}})$	$P_{\text{sat}}(T_{\text{LOX}})$
2	Ratio of initial LOX temperature to LOX saturation temperature corresponding to system (LOX) pressure, $T_{\text{LOX},i}/T_{\text{sat}}(P_{\text{LOX}})$	$P_{\text{sat}}(T_{\text{LOX}})$
3	Ratio of Helium volume flow to LOX volume, $\dot{V}_{\text{He}}/V_{\text{LOX}}$	$P_{\text{GOX},b}$
4	Ratio of Helium temperature to initial LOX temperature, $T_{\text{He}}/T_{\text{LOX},i}$	$P_{\text{GOX},b}$

Table 3
Parameter variations for test and analysis

No.	Independent parameters	Parameter variation range	Ref.
1	System pressure, P_{sys} (or P_{LOX}) (bar)	1, 3, 3.5 bar	
2	Initial LOX temperature, $T_{\text{LOX},i}$ (K)	91 K 99 K 102 K	Saturated at 1 bar Subcooled at 3.5 bar Saturated at 3 bar
3	Helium injection volume flow rate divided by LOX volume, $\dot{V}_{\text{He}}/V_{\text{LOX}}$ (min^{-1})	1–1.4 min^{-1}	
4	Injected helium temperature, T_{He} (K)	114, 140, 288 K	For 114 K, 140 K cases, helium is cooled by LN2 prior to injection

liquid oxygen temperature [10]. For atmospheric condition where liquid oxygen exists at saturated condition (90.2 K at 1 bar), ambient background heating loss was 150 W for the ambient-temperature-helium supply facility, Fig. 3, and 175 W for the low-temperature-helium supply facility, Fig. 4, respectively.

4.3. Parameters for analysis and tests

As mentioned above, the difference in the oxygen partial pressure in helium bubble $P_{\text{GOX},b}$ and the vapor pressure of liquid oxygen $P_{\text{sat}}(T_{\text{LOX}})$ is the main source of diffusion-driven evaporative cooling. Two methods can be conceived to increase the difference and the net diffusion-driven evaporation.

First method is to increase the vapor pressure. The maximum vapor pressure that liquid oxygen can attain, is the saturation vapor pressure corresponding to the saturation temperature of liquid oxygen. Since the vapor pressure increases with liquid oxygen temperature, saturated liquid oxygen is more susceptible to helium injection cooling than sub-cooled one.

Second method is to decrease the partial pressure of oxygen in helium bubble. To decrease the partial pressure of oxygen, high volume rate of helium is preferable. However, if the injected helium temperature is higher than liquid oxygen, the high mass flow inevitably accompanies high energy influx which vaporizes liquid oxygen by heat transfer between the warm helium and liquid oxygen. This heat induced vaporization subsequently increases gaseous oxygen partial pressure in helium bubbles. We have to consider, therefore, the net effect of helium flow rate. To

effectively lower the partial pressure of oxygen in helium bubble at high injected helium flow rate, low temperature helium can be utilized. Theoretically, if the injected helium has the same temperature with the liquid oxygen, the vaporization due to heat transfer is zero, and all the injected helium is used to lower the gaseous oxygen partial pressure in bubbles. Main parameters and their effects are qualitatively described in Table 2. The variation range of main parameters for experiments and analysis are established and summarized as in Table 3.

5. Results and discussions of analysis and experiments

5.1. Results for system pressure variation with ambient temperature helium injection

The comparison of measured cooling performances for two system pressures along with analytical results is shown in Fig. 5: atmospheric and pressurized condition (3 bar). Ambient temperature helium (288 K) is injected into liquid oxygen for both cases. For atmospheric system pressure, the liquid oxygen is cooled from its initially saturation temperature at 91 K down to 85.3 K after 1500 s with 100 l/min ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.39 \text{ min}^{-1}$) helium injection rate. For pressurized condition case (at 3 bar), on the other hand, the initially sub-cooled liquid oxygen (91 K) is warmed during helium injection with 87 l/min ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.21 \text{ min}^{-1}$) injection rate.

Liquid oxygen temperature variation is directly influenced by the vaporization rate of liquid oxygen. The vaporization rate is analyzed into two components. The first one is the vaporization due to heat transfer from helium gas (vapor side heat transfer) which is the first term on the right

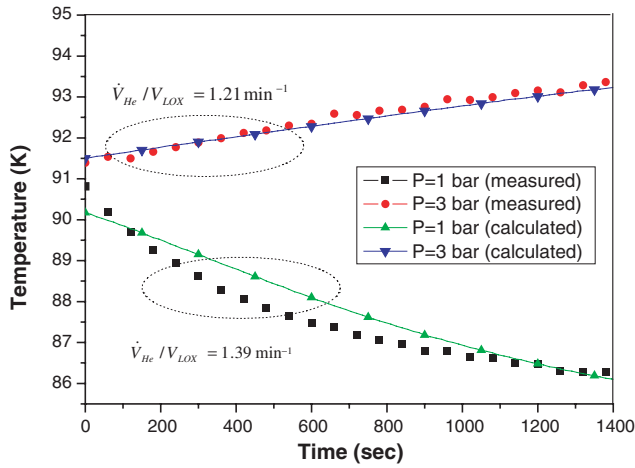


Fig. 5. Comparison of cooling performance between atmospheric and pressurized condition ($T_{\text{LOX},i} = 91 \text{ K}$, $T_{\text{He}} = 288 \text{ K}$).

in Eq. (4), defined as $A_s(h_{\text{gi}}(T_g - T_i))/h_{\text{fg}}$. The second one is the vaporization due to heat transfer from liquid oxygen (liquid side heat transfer) which is the second term on the right in Eq. (4), defined as $A_s(-h_{\text{li}}(T_i - T_l))/h_{\text{fg}}$, accounting for diffusion-driven evaporation. The interface temperature T_i is calculated based on the instantaneous partial pressure of gaseous oxygen in the gas region, $T_i = T_{\text{sat}}(P_{\text{GOX}})$. The two vaporization components are coupled as once the vaporization due to the vapor side heat transfer increases oxygen partial pressure in the gas region, the liquid side heat transfer driven by mass diffusion decreases. The reference analysis condition is that 90 l liquid oxygen (91 K) is filled and the helium gas with 90 l/min ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1 \text{ min}^{-1}$) at 288 K is injected into liquid oxygen. The result in Fig. 6 shows that for the case of 3 bar system pressure, the vaporization by the vapor side contribution is dominant compared to the liquid side heat transfer component. The liquid side heat transfer actually has small negative value indicating condensation occurs instead

of vaporization. Whereas, for the case of 1 bar system pressure, the vaporization by the vapor side heat transfer is smaller than the vaporization by the liquid side. It can be explained that for pressurized condition (3 bar), about 3 times more helium mass flow is injected into the liquid oxygen compared to the atmospheric case with the same volume flow rate, which greatly increases vaporization due to heat transfer between the warm helium and liquid oxygen. This situation increases gaseous oxygen partial pressure in helium bubble and subsequently suppresses diffusion-driven evaporation.

Fig. 7 shows the comparison of gaseous oxygen partial pressure and corresponding liquid oxygen vapor pressure for the two cases. For the atmospheric pressure system, the gaseous oxygen partial pressure is lower than liquid oxygen saturation pressure, however, for pressurized condition (3 bar), the gaseous oxygen partial pressure is

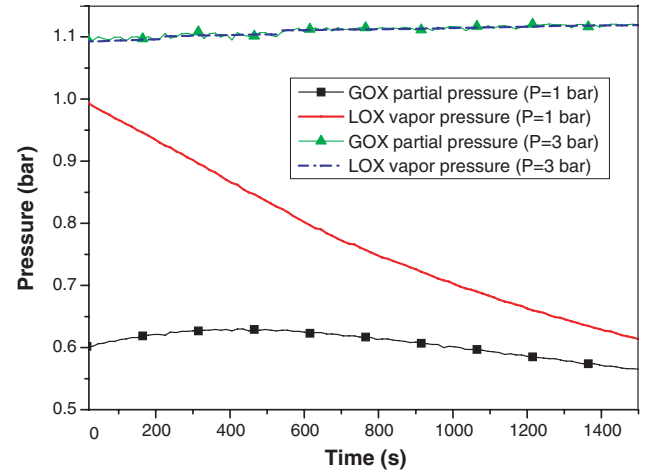


Fig. 7. Comparison of GOX partial pressure and LOX vapor pressure between atmospheric and pressurized condition ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1 \text{ min}^{-1}$, $T_{\text{LOX},i} = 91 \text{ K}$, $T_{\text{He}} = 288 \text{ K}$).

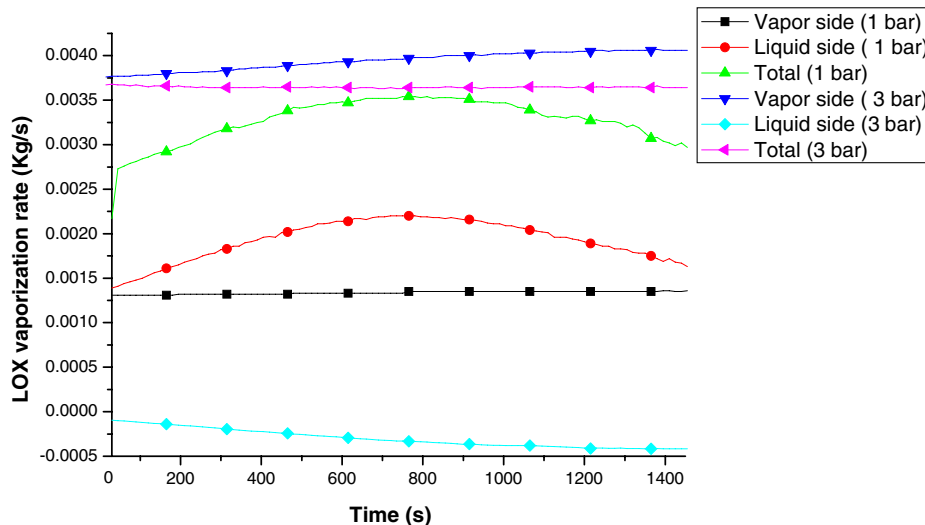


Fig. 6. Comparison of vaporization components between atmospheric and pressurized condition ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1 \text{ min}^{-1}$, $T_{\text{LOX},i} = 91 \text{ K}$, $T_{\text{He}} = 288 \text{ K}$).

slightly higher than liquid oxygen saturation pressure which suggests that condensation occurs and subsequent warming of liquid oxygen.

For pressurized case, the gaseous oxygen partial pressure in Fig. 7 and the corresponding vaporization by the vapor side heat transfer in Fig. 6 indicate the cause of failure of liquid oxygen cooling by helium injection.

5.2. Results for initial liquid oxygen temperature variation

The comparison of experimental cooling performances for two liquid oxygen initial temperatures along with analytical results under pressurized conditions (3 bar) is shown in Fig. 8: 91 K sub-cooled liquid oxygen and 102 K saturated liquid oxygen. For saturated case, the test was initiated when liquid oxygen, due to ambient heating, reached its saturation temperature. Ambient temperature helium (288 K) is injected into liquid oxygen for both cases with varying injection flow rates. The experimental results give distinctive difference between the initially sub-cooled and the initially saturated liquid oxygen under the helium injection process. For initially saturated liquid oxygen, the liquid oxygen is cooled from its saturation temperature of 102 K at 3 bar down to 94 K after 1500 s with 96 l/min ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.34 \text{ min}^{-1}$) injection rate. For the initially sub-cooled case ($T_{\text{LOX},i} = 91 \text{ K}$), helium injection induces heating. The different performance between the two cases can be explained by the difference of the vapor pressure of liquid oxygen because the vaporization rate by vapor side heat transfer is almost same. The vapor pressure of 102 K liquid oxygen is 3 bar and the vapor pressure of 91 K liquid oxygen is 1 bar. Since higher liquid oxygen initial temperature creates higher liquid oxygen vapor pressure, the difference between the liquid oxygen vapor pressure and the oxygen partial pressure in the helium bubble ($P_{\text{GOX},b}$) increases with liquid oxygen initial temperature, which enhances diffusion-driven evaporative cooling.

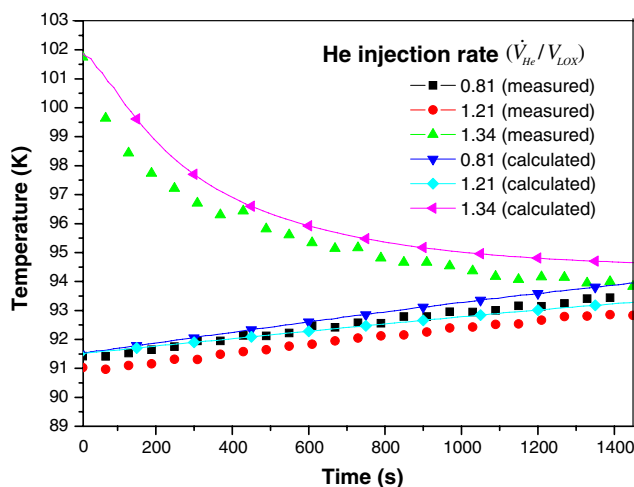


Fig. 8. Comparison of cooling performance between sub-cooled LOX and saturated LOX ($P = 3 \text{ bar}$, $T_{\text{He}} = 288 \text{ K}$).

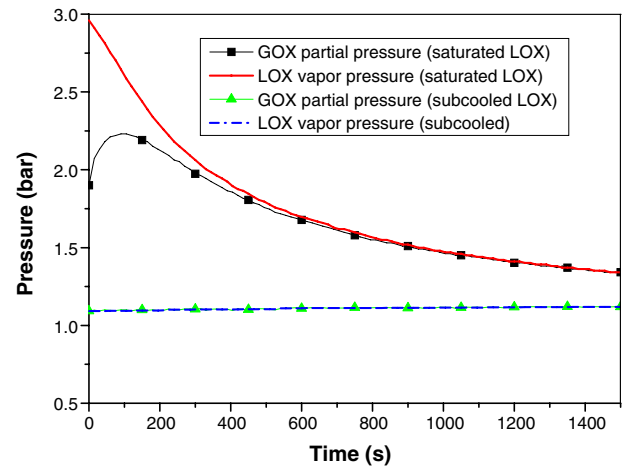


Fig. 9. Comparison of GOX partial pressure and LOX vapor pressure between sub-cooled LOX and saturated LOX ($P = 3 \text{ bar}$, $T_{\text{He}} = 288 \text{ K}$).

Fig. 9 shows the comparison of gaseous oxygen partial pressure and corresponding liquid oxygen vapor pressure for the two cases. For initially saturated liquid oxygen, the vapor pressure is so large that the developed gaseous oxygen partial pressure of even more than 2 bar cannot still stop diffusion-driven evaporative cooling. It is concluded that saturated liquid oxygen has always an advantage over the sub-cooled liquid oxygen for the purpose of helium injection cooling. Increasing the helium flow rate for the sub-cooled liquid oxygen suppresses the warming rate but cannot reverse the warming tendency as shown in Fig. 8.

5.3. Results for helium temperature variation

To examine the effect of injected helium temperature, preliminary tests were performed under atmospheric condition with helium temperature variation. Fig. 10 shows the experimental results for saturated liquid oxygen under atmospheric condition. Two cases are noteworthy: 288 K

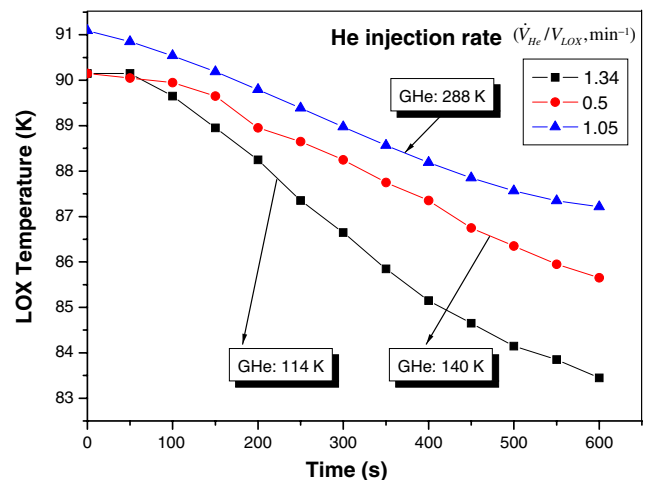


Fig. 10. Comparison of cooling performance for helium temperature variation ($P = 1 \text{ bar}$).

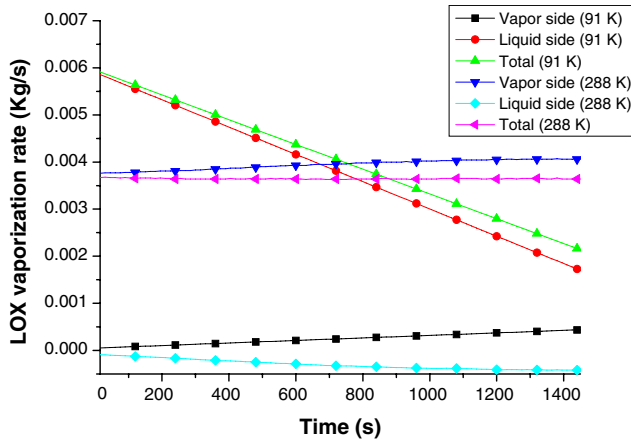


Fig. 11. Comparison of vaporization components between two helium temperatures ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1 \text{ min}^{-1}$, $T_{\text{LOX},i} = 91 \text{ K}$, $P = 3 \text{ bar}$).

injection and 140 K injection. For 140 K injection, although helium injection rate ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 0.5 \text{ min}^{-1}$) is lower than 288 K injection ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.34 \text{ min}^{-1}$), more sub-cooling is obtained, which indicates that injection of low temperature helium is an effective means for helium injection cooling purpose. For the colder 114 K injection, we get more sub-cooling effect than that of 140 K injection case.

Fig. 11 shows the analysis results for comparison of the vaporization components under pressurized condition (3 bar) between two helium temperatures: 288 K helium injection and 91 K helium injection. For the case of 91 K helium injection, the vapor side contribution is much lower than in the 288 K case, but due to the small partial pressure of the gaseous oxygen, the liquid side contribution is larger. It is also noticeable that the vaporization due to the vapor side heat transfer can increase oxygen partial pressure in the gas region, which subsequently decreases the liquid side heat transfer driven by mass diffusion. Therefore, the vaporization rate ultimately decreases.

5.4. Results for sub-cooled pressurized conditions with helium temperature variation

Fig. 12 shows the experimental results for comparison of sub-cooled pressurized conditions with helium temperature variation. One is initially 99 K sub-cooled liquid oxygen at 3.5 bar (the saturation temperature at 3.5 bar is about 104 K) into which low temperature (114 K) helium is injected ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.3 \text{ min}^{-1}$). The other is initially 91 K sub-cooled liquid oxygen at 3 bar into which atmospheric temperature (288 K) helium is injected ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.21 \text{ min}^{-1}$). The result shows that only for the initially 99 K sub-cooled LOX, the temperature decreases. Cooling occurs down to 86.8 K at 1500 s by low temperature (114 K) helium injection.

Fig. 13 shows the analysis results for comparison of the vaporization components between two cases under 3.5 bar and specified helium flow rate ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1 \text{ min}^{-1}$). The initial liquid oxygen temperatures are specified as identical

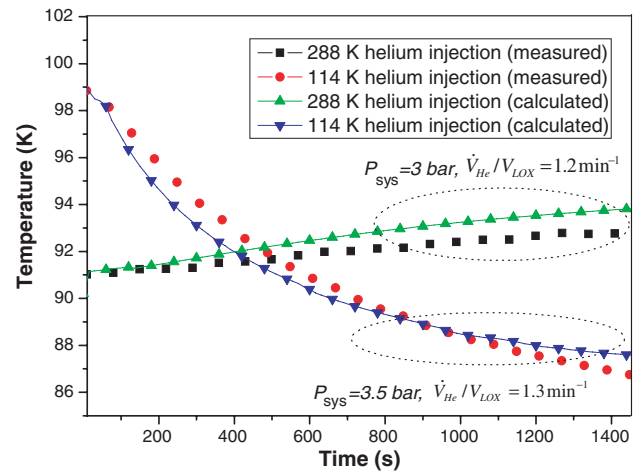


Fig. 12. Comparison of sub-cooled and pressurized conditions with helium temperature variation.

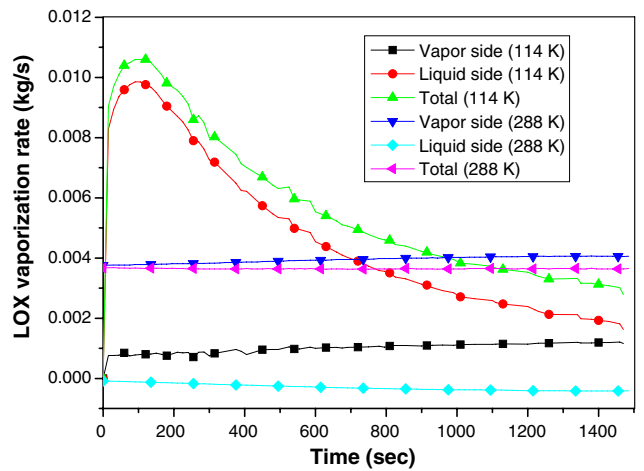


Fig. 13. Comparison of vaporization components between two sub-cooled and pressurized conditions with helium temperature variation ($P = 3.5 \text{ bar}$, $\dot{V}_{\text{He}}/V_{\text{LOX}} = 1 \text{ min}^{-1}$).

values as the test conditions (99 K and 91 K). For the low temperature helium injection case, the vaporization rate due to vapor side heat transfer is relatively small compared to that of the ambient temperature injection case, which keeps gaseous oxygen partial pressure below the saturation pressure of liquid oxygen as shown in Fig. 14. From these experimental and analytical results, we can conclude the injection of low temperature helium induces cooling of sub-cooled liquid oxygen even under pressurized condition.

5.5. Summary of the results

Several observations presented above have shown that we can divide the cases of liquid oxygen cooling by helium injection into two categories according to initial liquid oxygen temperature: saturated or sub-cooled. For initially saturated liquid oxygen, sub-cooling of liquid oxygen is easily achieved regardless of injected helium temperature

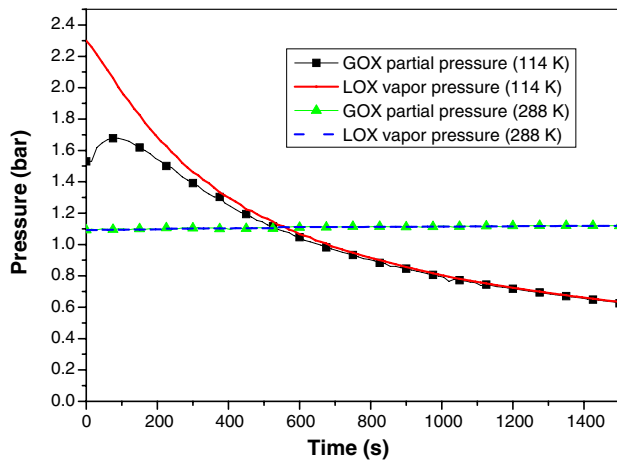


Fig. 14. Comparison of GOX partial pressure and LOX vapor pressure between two sub-cooled and pressurized conditions with helium temperature variation ($P = 3.5$ bar, $\dot{V}_{\text{He}}/\dot{V}_{\text{LOX}} = 1 \text{ min}^{-1}$).

even under pressurized condition. However, for initially sub-cooled liquid oxygen, the situation is a little bit complex. If the generated oxygen partial pressure in bubbles exceeds the vapor pressure of liquid oxygen, the vaporization induced by diffusion does not occur. Therefore, in order to cool the initially sub-cooled liquid oxygen, gaseous oxygen partial pressure should be maintained as low as possible. The method of lowering the injected helium temperature turns out to be effective for cooling purpose.

6. Simplified model for determining cooling condition

6.1. General aspects of model simplification

As seen from the above discussion, if the gaseous oxygen partial pressure in helium bubble is lower than the liquid oxygen vapor pressure, cooling occurs due to diffusion mass transfer. The difference between the gaseous oxygen partial pressure and liquid oxygen vapor pressure is the largest at initial injection stage and reduces as the sub-cooling of LOX proceeds, which accompanies decrease of liquid oxygen vapor pressure and mixture temperature. The gaseous oxygen partial pressure developed in initial injection stage should be below the corresponding liquid oxygen vapor pressure for cooling liquid oxygen.

In the following sections, we develop a simplified model for predicting cooling condition. The core of this model is to estimate the gaseous oxygen partial pressure that is developed during initial helium injection stage and to identify the dominant operating parameters as non-dimensional form to describe system behavior generally with different configurations and operating conditions. The model assumes gaseous oxygen partial pressure is developed by heat transfer only between helium gas and liquid oxygen in advance, and then liquid oxygen vaporization due to diffusion occurs with reference to the gaseous oxygen partial pressure developed

by heat transfer. This simplification allows us to investigate the effect of gaseous oxygen partial pressure developed by the evaporation due to diffusion. The model may be useful for designers to obtain some rough combination of important parameters which enables liquid oxygen cooling without resorting to extensive numerical analysis during preliminary design stage of liquid propellant rocket.

6.2. Liquid oxygen vaporization rate due to heat transfer at initial injection stage

Liquid oxygen vaporization is assumed to be developed from heat transfer between injected helium bubbles and liquid oxygen at the initial injection stage. For calculating the heat transfer rate, the net bubble surface area and the heat transfer coefficient are required.

Single bubble surface area at the initial injection stage may be calculated from the equivalent detached bubble size from the submerged nozzle in liquid oxygen, and the rising velocity of the bubble is calculated based on that detached bubble size. The correlations for single bubble size at detachment and corresponding rise velocity can be found from the references with the constant injection flow rate $\dot{V}_{\text{He}}$ and liquid and gas properties [6,11]. Selection criterion of the proper model is also discussed in the references. For the conditions of the investigations in this paper, Eq. (8), which describes the volume of a bubble at detachment in an inviscid liquid [11], is used for the estimation of the detached bubble volume $V_{b,d}$ and Eq. (9), which describes the rising velocity for a bubble with diameter greater than 5 mm is used for the corresponding bubble rising velocity u_b [12]

$$V_{b,d} = 1.138 \frac{\dot{V}_{\text{He}}^{6/5}}{g^{3/5}}, \quad (8)$$

$$u_b = 1.05 \left[\frac{g(\rho_l - \rho_g)}{\rho_l} \cdot \left(\frac{D_{b,d}}{2} \right) \right]^{1/2}. \quad (9)$$

The equivalent detached bubble diameter $D_{b,d}$ and the corresponding surface area of the detached bubble $A_{b,d}$ are calculated by Eq. (10) as a spherical bubble

$$D_{b,d} = \left(\frac{6V_{b,d}}{\pi} \right)^{1/3}, \quad A_{b,d} = \pi D_{b,d}^2. \quad (10)$$

The bubble formation frequency f_b (i.e., the number of bubbles formed at the orifice per unit time) is determined as [13]

$$f_b = \frac{\dot{V}_{\text{He}}}{V_{b,d}}, \quad (11)$$

where $\dot{V}_{\text{He}}$ is the volume flow rate of helium through orifice and $V_{b,d}$ is the detached bubble volume, respectively. If we define f'_b as the number of bubbles generated during initial 1 s ($f'_b \cdot 1 \text{ s}$), the averaged net surface area of the generated bubbles during the initial 1 s, can be expressed

$$A_s = \frac{1}{2} f'_b A_{b,d}. \quad (12)$$

The heat transfer from helium gas to the interface of liquid oxygen is in the form of natural convection inside spherical bubble. Empirical natural convection correlation proposed by Kreith [14] is applied for calculating the heat transfer coefficient as shown in Eq. (13), inside spherical cavity, which is known to be applicable to small bubble with low Rayleigh number ($10^4 < Ra \leq 10^9$)

$$\overline{Nu}_D = 0.59 Ra_D^{1/4}, \quad (13)$$

$$Ra_D = \frac{g\beta(T_g - T_w)D^3}{\alpha\nu},$$

where the injected helium temperature, T_{He} is used for the gas temperature, T_g . Since the bubble temperature change may be small during the initial injection stage (initial 1 s), $T_{He} \approx T_g$ and the liquid oxygen saturation temperature $T_{sat}(P_{LOX})$ is used for the wall temperature (T_w), respectively. The detached bubble diameter $D_{b,d}$ is used for the cavity diameter, D . The coefficient of expansion, β , kinematic viscosity, ν and thermal diffusivity, α are evaluated at the film temperature $(T_{He} + T_{sat}(P_{LOX}))/2$.

Using the specified helium volume flow rate, $\dot{V}_{He}$, helium mass flow rate, $\dot{m}_{He}$ is simply expressed as follows:

$$\dot{m}_{He} = \rho_{He} \dot{V}_{He} \quad (14)$$

The liquid oxygen vaporization rate due to heat transfer ($\dot{m}_{LOX,h}$) at the initial injection stage is obtained by balancing the energy required for vaporization with the heat transfer from helium gas to liquid oxygen in bubbles as follows:

$$\dot{m}_{LOX,h} = A_s (h_{gi}(T_{He} - T_{sat}(P_{LOX}))/h_{fg}), \quad (15)$$

where A_s is the average surface area of the generated bubbles obtained from Eq. (12), h_{gi} is the heat transfer coefficient between the helium bubble and the liquid oxygen from Eq. (13), h_{fg} is the heat of vaporization of liquid oxygen at the LOX initial temperature. Comparing Eq. (15) with the first term on the right of Eq. (4), the gas temperature T_g in Eq. (4) is set equal to the helium injection temperature T_{He} only for the initial injection stage. For the interfacial temperature T_i , $T_{sat}(P_{LOX})$, the saturation temperature of liquid oxygen, is applied, which is the equilibrium temperature at the interface of volatile liquid. This reasoning is valid as long as the mass transfer of oxygen vapor into helium bubbles is sufficiently small [10]. Since the analysis is focused on oxygen vapor generation rate without considering the mass transfer rate into bubbles, the equilibrium condition can be applied for the interfacial temperature.

6.3. Gaseous oxygen partial pressure in bubbles at initial injection stage

Assuming now the generated oxygen vapor is propelled instantaneously into the helium bubbles without degrada-

tion (not diffusion limited), the instantaneous equilibrium mole ratio X is defined by

$$X = \frac{n_{He,b}}{n_{GOX,b}}, \quad (16)$$

where $n_{He,b}$ and $n_{GOX,b}$ are the moles of helium gas and the moles of oxygen vapor in the bubble, respectively. The partial pressure of oxygen or helium constituent is assumed to satisfy the real gas law as shown in Eq. (17)

$$PV = ZnRT. \quad (17)$$

Neglecting the pressure difference due to the surface tension in a bubble ($2\sigma/r_b$), according to Dalton's law, the pressure relation requires

$$P_{He,b} = P_{sys} - P_{GOX,b}. \quad (18)$$

Substituting Eqs. (17) and (18) into Eq. (16), the mole ratio parameter X , then can be related to the partial pressure ratio as follows:

$$X = \frac{P_{sys} - P_{GOX,b}}{P_{GOX,b}} \cdot \frac{Z_{GOX}}{Z_{GHe}}, \quad (19)$$

which suggests that if the equilibrium mole ratio parameter X at initial injection stage is known, the gaseous oxygen partial pressure $P_{GOX,b}$ can be obtained with the given system pressure P_{sys} and the compressibility factor ratio Z_{GOX}/Z_{GHe} .

Using Eq. (19), the gaseous oxygen partial pressure in the bubble $P_{GOX,b}$ is expressed with the calculated non-dimensional parameter X as in Eq. (20)

$$P_{GOX,b} = \frac{Z_{GOX}/Z_{GHe}}{X + (Z_{GOX}/Z_{GHe})} \cdot P_{sys}. \quad (20)$$

For initially saturated liquid oxygen, system pressure P_{sys} can be replaced by the saturation pressure of liquid oxygen $P_{sat}(T_{LOX})$. Since the mole ratio X has a positive value, it is evident that the cooling requirement ($P_{GOX,b} < P_{sat}(T_{LOX})$) is always satisfied regardless of the system pressure and the injected helium temperature. This reasoning agrees with the test results for initially saturated LOX presented in the previous chapter. Whereas, for initially sub-cooled LOX, $P_{sat}(T_{LOX})$ is less than P_{sys} . Therefore, $P_{GOX,b}$ should be less than $P_{sat}(T_{LOX})$ for cooling by selecting a proper value of X . The cooling limit condition is set when $P_{GOX,b}$ is equal to $P_{sat}(T_{LOX})$.

From Eq. (16), the mole ratio change rate can be calculated as follows:

$$\frac{\dot{n}_{GOX}}{\dot{n}_{He}} = \frac{1}{X} - \frac{\dot{X}}{X^2} \cdot \frac{n_{He}}{\dot{n}_{He}}. \quad (21)$$

Assuming the instantaneous equilibrium is obtained, $\dot{X}$ can be thought as zero and Eq. (21) reduces to

$$X = \frac{\dot{n}_{He}}{\dot{n}_{GOX}}. \quad (22)$$

Since the number of moles of each component is equal to the mass of that component divided by its molar mass as shown in Eq. (23), the mole ratio parameter X can be described in terms of mass flow rate ratio as shown in Eq. (24)

$$\dot{n}_{\text{He}} = \dot{m}_{\text{He}}/M_{\text{He}}, \quad \dot{n}_{\text{GOX}} = \dot{m}_{\text{GOX}}/M_{\text{GOX}}, \quad (23)$$

$$X = \frac{\dot{n}_{\text{GHe}}}{\dot{n}_{\text{GOX}}} = \frac{\dot{m}_{\text{He}}M_{\text{GOX}}}{\dot{m}_{\text{GOX}}M_{\text{He}}}. \quad (24)$$

Eq. (24) allows us to estimate the mole rate ratio X with the given helium mass flow rate $\dot{m}_{\text{He}}$ and the LOX vaporization rate due to heat transfer ($\dot{m}_{\text{GOX}} = \dot{m}_{\text{LOX,h}}$) at the initial injection stage.

By making use of Eqs. (12), (14) and (15), the mole ratio parameter X for initial helium injection stage can be finally expressed as follows:

$$X = \frac{\dot{m}_{\text{He}}M_{\text{GOX}}}{\dot{m}_{\text{GOX}}M_{\text{He}}} = \frac{2\rho_{\text{He}}\dot{V}_{\text{He}}h_{\text{fg}}}{h_{\text{gi}}f'_{\text{b}}A_{\text{b,d}}(T_{\text{He}} - T_{\text{sat}})} \frac{M_{\text{GOX}}}{M_{\text{GHe}}} \quad (f'_{\text{b}} = \dot{V}_{\text{He}}/V_{\text{b,d}} \cdot 1 \text{ s}) \quad (25)$$

6.4. Liquid oxygen cooling rate at the initial injection stage

In determining cooling condition, the energy balance equation, Eq. (2) is carefully re-investigated. First, note that the ambient heat flux $\dot{Q}_{\text{amb}}$ in the equation tends to increase the temperature of liquid oxygen. We should notice that the higher liquid oxygen temperature caused by $\dot{Q}_{\text{amb}}$ creates the higher liquid oxygen vapor pressure. As discussed in the preceding chapters, the difference in the vapor pressure of liquid oxygen and the partial pressure of oxygen gas in helium bubble is the main factor of diffusion-driven evaporative cooling (helium injection cooling). Since the higher liquid oxygen temperature caused by $\dot{Q}_{\text{amb}}$ creates higher liquid oxygen vapor pressure, it enhances diffusion-driven evaporative cooling. From this point of view, smaller $\dot{Q}_{\text{amb}}$ gives more severe situation for cooling condition. To the designers who are interested in combination of parameters adequate for cooling, the ability to predict cooling condition in most severe condition ($\dot{Q}_{\text{amb}} = 0$) is of considerable value especially in preliminary design stage. Therefore, we set $\dot{Q}_{\text{amb}}$ as zero in this work.

Second, note that the heat transfer rate from the gas to the liquid interface can be expressed in terms of vaporization due to heat transfer as shown in Eq. (26)

$$\dot{Q}_{\text{gi}} = A_s(h_{\text{gi}}(T_{\text{g}} - T_{\text{i}})) \approx \dot{m}_{\text{LOX,h}}h_{\text{fg}}. \quad (26)$$

By making use of Eqs. (4) and (26), Eq. (2) with $\dot{Q}_{\text{amb}} = 0$, we can reduce the rate at which the bulk liquid oxygen energy changes as follows:

$$m_{\text{LOX}}C_{p,\text{LOX}} \frac{dT_{\text{LOX}}}{dt} + A_sh_{\text{li}}(T_{\text{LOX}} - T_{\text{i}}) = 0, \quad (27)$$

where the interface temperature T_{i} is the calculated liquid oxygen saturation temperature based on the gaseous oxygen partial pressure in bubbles using finite heat transfer and instantaneous diffusional mass transfer model as

shown in Eq. (5), which accounts for the effect of diffusion-driven evaporative cooling [6]. With the assumption of instantaneous diffusion and thermodynamic phase equilibrium of oxygen vapor and helium gas at the interface, the interface temperature T_{i} in Eq. (27) can be related to the gaseous oxygen partial pressure through Clausius–Clapeyron equation for an ideal gas, Eq. (28),

$$T_{\text{i}} = T_{\text{sat}}(P_{\text{GOX,b}}) = \frac{H_{\text{fg}}T_{\text{ref}}}{H_{\text{fg}} - T_{\text{ref}}R \cdot \ln(P_{\text{GOX,b}}/P_{\text{ref}})}, \quad (28)$$

where H_{fg} is the vaporization heat of liquid oxygen per mole, $P_{\text{GOX,b}}$ is the partial pressure of oxygen vapor in the helium bubble, R is the universal gas constant ($8.3145 \text{ J mol}^{-1} \text{ K}^{-1}$) and T_{ref} is the normal boiling-point temperature at atmospheric pressure P_{ref} . Eq. (27) is solved for liquid (liquid oxygen) cooling rate dT_{LOX}/dt and liquid oxygen temperature T_{LOX} is set equal to its initial temperature $T_{\text{LOX,i}}$, then temperature change rate of liquid oxygen during the initial injection stage is approximated as follows:

$$\frac{\Delta T_{\text{LOX}}}{\Delta t} = -\frac{A_sh_{\text{li}}}{m_{\text{LOX}}C_{p,\text{LOX}}} \cdot (T_{\text{LOX,i}} - T_{\text{i}}), \quad (29)$$

where h_{li} is the boiling heat transfer coefficient associated with liquid oxygen vaporization in the ‘u–i’ region as shown in Fig. 2. This calculation is calculated based on the temperature difference between liquid oxygen, T_{LOX} and the corresponding saturation temperature of liquid oxygen, $T_{\text{sat}}(P_{\text{LOX}})$. The heat transfer coefficient, h_{li} does not affect whether liquid oxygen is cooled or not, and only influences the absolute value of the temperature change rate. The detailed description and correlations for h_{li} is found elsewhere [6].

Since T_{i} is associated with the partial pressure of oxygen vapor developed by heat transfer, Eq. (29) allows us to calculate liquid oxygen cooling rate (or warming rate) due to diffusion-driven evaporation (or condensation) with respect to the gaseous oxygen partial pressure developed by the heat transfer during the initial injection stage. In Eq. (29), for cooling liquid oxygen, it is necessary for the interface temperature T_{i} to be lower than the liquid oxygen initial temperature, $T_{\text{LOX,i}}$, i.e., $T_{\text{LOX,i}} > T_{\text{i}}$.

Substituting $P_{\text{GOX,b}}$ in Eq. (20) to calculate T_{i} , Eq. (28), and then substituting T_{i} into the cooling requirement ($T_{\text{LOX,i}} > T_{\text{i}}$) in Eq. (29) gives the cooling condition after some manipulations

$$X > \left(\frac{Z_{\text{GOX}}}{Z_{\text{GHe}}} \right) \cdot \left[\frac{P_{\text{sys}}}{P_{\text{ref}}} \cdot e^{\frac{H_{\text{fg}}}{T_{\text{ref}}R} \left(\frac{T_{\text{ref}}}{T_{\text{LOX,i}}} - 1 \right)} - 1 \right]. \quad (30)$$

Eq. (30) indicates that the non-dimensional parameter X which is the combination of bubble characteristic parameters and bubbling system operating parameters, should be established to be higher than the value of the right hand side of Eq. (30) which can be defined as the critical $X(X_{\text{cr}})$ for obtaining liquid oxygen cooling.

Substituting X expression, Eq. (25) into Eq. (30) gives the following helium temperature condition for cooling:

$$T_{\text{He}} < \frac{2\rho_{\text{He}}V_{\text{LOX}}}{(f'_{\text{b}}A_{\text{b,d}}/h_{\text{fg}}) \cdot h_{\text{gi}}} \cdot \left(\frac{\dot{V}_{\text{He}}}{V_{\text{LOX}}} \right) \cdot \left(\frac{M_{\text{GOX}}}{M_{\text{He}}} \right) \cdot \left[\frac{\frac{Z_{\text{He}}}{Z_{\text{GOX}}}}{\frac{P_{\text{sys}}}{P_{\text{ref}}} \cdot e^{\frac{H_{\text{fg}}}{T_{\text{ref}}R} \left(\frac{T_{\text{ref}}}{T_{\text{LOX,i}}} - 1 \right)} - 1} \right] + T_{\text{sat}}(P_{\text{LOX}}). \quad (31)$$

$(f'_{\text{b}} = \dot{V}_{\text{He}}/V_{\text{b,d}} \cdot 1 \text{ s})$

Eq. (31) should be solved iteratively as ρ_{He} , A_{s} , h_{gi} , Z_{He} depend on injected helium temperature T_{He} . Since $\dot{V}_{\text{He}}/V_{\text{LOX}}$ usually has a limited range of variation among systems, Eq. (31) can be used to calculate the maximum allowable helium temperature (critical helium temperature, $T_{\text{He,crit}}$) with the specified volume flow rate for cooling sub-cooled pressurized liquid oxygen.

6.5. Calculation results

Table 4 shows the calculation results by the simplified model introduced in the previous sections. The calculations are performed for the representative test and analysis cases presented in this paper.

The following eight cases are analyzed: (1) initially saturated liquid oxygen at 1 bar into which ambient temperature helium (288 K) is injected (with gas flow rate, $\dot{V}_{\text{He}}/V_{\text{LOX}} = 1 \text{ min}^{-1}$), (2) initially saturated liquid oxygen at 1 bar into which ambient temperature helium (288 K) is injected with the increased gas flow rate ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.3 \text{ min}^{-1}$), (3) initially saturated liquid oxygen at 1 bar into which low temperature helium (114 K) instead of ambient

temperature helium is injected with the same gas flow rate ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.3 \text{ min}^{-1}$), (4) initially saturated liquid oxygen at 3 bar into which ambient temperature helium (288 K) is injected with large gas flow rate ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.3 \text{ min}^{-1}$), (5) initially sub-cooled liquid oxygen at 3 bar into which ambient temperature helium (288 K) is injected with small gas flow rate ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 0.8 \text{ min}^{-1}$), (6) initially sub-cooled liquid oxygen at 3 bar into which ambient temperature helium (288 K) is injected with the increased gas flow rate ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.2 \text{ min}^{-1}$), (7) initially sub-cooled liquid oxygen (98 K) at 3.5 bar into which low temperature helium (114 K) is injected (with the gas flow rate $\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.3 \text{ min}^{-1}$), (8) initially sub-cooled liquid oxygen (91 K) at 3.5 bar into which low temperature helium (114 K) is injected (with the gas flow rate $\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.0 \text{ min}^{-1}$). Table 4 also shows the net surface area of bubbles (calculated from Eq. (11)) and the heat transfer coefficients calculated from the detached bubble analysis discussed in the previous sections for the individual cases. For pressurized condition, the gas-to-interface heat transfer coefficient increases due to higher helium density than that of atmospheric condition.

The results show that the calculated gaseous oxygen partial pressures are lower than the liquid oxygen vapor pressures for the cooling conditions (cases 1, 2, 3, 4, 7, 8). However, for the warming condition (cases 5, 6), the tendency is reversed. The calculation results agree with the experimental observation and the analytical results presented in the previous chapter. Therefore, the parameter, X , can be regarded as a useful guide for determining cooling condition during preliminary design stage of helium injection cooling system.

Fig. 15 shows the graphical representation of the relation between the parameter X and the cooling rate for

Table 4
Calculation results of simplified model for initial injection stage

Parameters	Case No.							
	1	2	3	4	5	6	7	8
Initial LOX temperature, $T_{\text{LOX,i}}$ (K)	91 (saturated)	91 (saturated)	91 (saturated)	102 (saturated)	91 (subcooled)	91 (subcooled)	98 (subcooled)	91 (subcooled)
System pressure, P_{sys} (or P_{LOX}) (bar)	1	1	1	3	3	3	3.5	3.5
Injected helium flow rate, $\dot{V}_{\text{He}}/V_{\text{LOX}}$ (min^{-1})	1.0	1.3	1.3	1.3	0.8	1.3	1.3	1.0
Injected helium temperature, T_{He} (K)	288	288	114	288	288	288	114	114
Diameter of detached bubble, $D_{\text{b,d}}$ (m)	0.052	0.058	0.046	0.058	0.047	0.058	0.046	0.042
Rising velocity of detached bubble, u_{b} (m/s)	0.529	0.558	0.501	0.558	0.506	0.558	0.474	0.474
Net surface area of bubbles, A_{s} (m^2)	0.058	0.068	0.049	0.071	0.051	0.065	0.049	0.041
Gas-to-interface heat transfer coefficient, h_{gi} ($\text{W}/\text{m}^2 \text{ K}$)	30.0	29.2	23.2	49.2	52.4	50.0	39.3	43.5
Liquid-to-interface heat transfer coefficient, h_{li} ($\text{W}/\text{m}^2 \text{ K}$)	2573	2561	2585	3033	1327	1345	1585	1322
Mole fraction ratio, X	0.75	0.86	18.81	1.56	1.23	1.57	89.9	73.1
GOX partial pressure, $P_{\text{GOX,b}}$ (bar)	0.57	0.54	0.05	1.16	1.34	1.17	0.039	0.047
LOX vapor pressure, $P_{\text{sat}}(T_{\text{LOX,i}})$ (bar)	1.0	1.0	1.0	3.0	1.0	1.0	2.3	1.0
Interface temperature, T_{i} (K)	84.2	83.6	65.9	91.6	93.4	91.7	64.3	65.5
Cooling rate, $\Delta T_{\text{LOX}}/\Delta t$ (K/s)	−0.0078	−0.0098	−0.046	−0.018	0.0018	0.0013	−0.038	−0.021

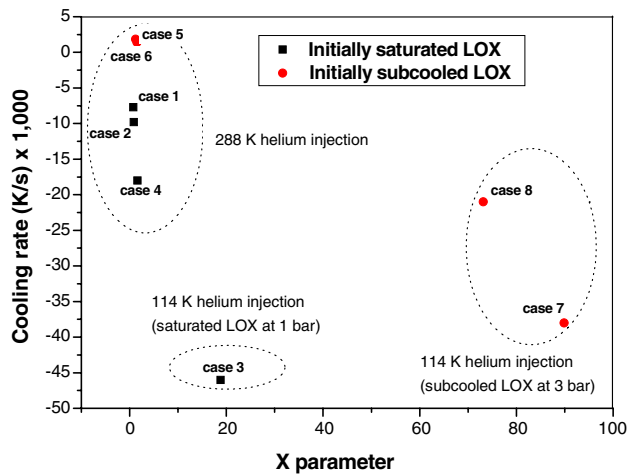


Fig. 15. Graphical representation of the relation between X parameter and cooling rate.

the cases mentioned above. For ambient temperature helium (288 K) injection, with similar X value, cooling or warming occurs according to the liquid oxygen initial condition: sub-cooled or saturated.

For low temperature helium (114 K) injection cases, the X parameters are large mainly due to the decrease of heat induced vaporization and the increased helium density. This higher X value subsequently increases the cooling rate of liquid oxygen.

In addition, among the low temperature helium (114 K) injection cases, the case 7 and the case 8 (sub-cooled at 3.5 bar) show a much lower sensitivity with respect to cooling rate than the case 3 (saturated at atmospheric system pressure). That is mainly because for the pressurized cases, boiling is suppressed, therefore, liquid-to-interface boiling heat transfer coefficient h_{li} is smaller than atmospheric case as shown in Table 4.

Fig. 16 shows the calculation results for critical X value, X_{cr} , defined as the minimum value of X for cooling liquid

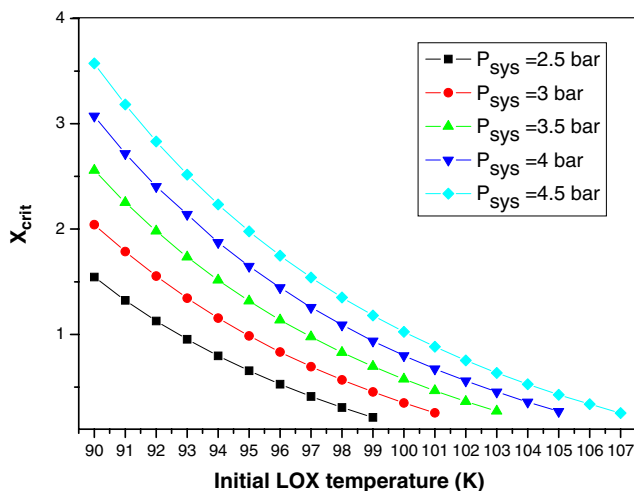


Fig. 16. Critical X parameter with initial LOX temperature variation under pressurized conditions.

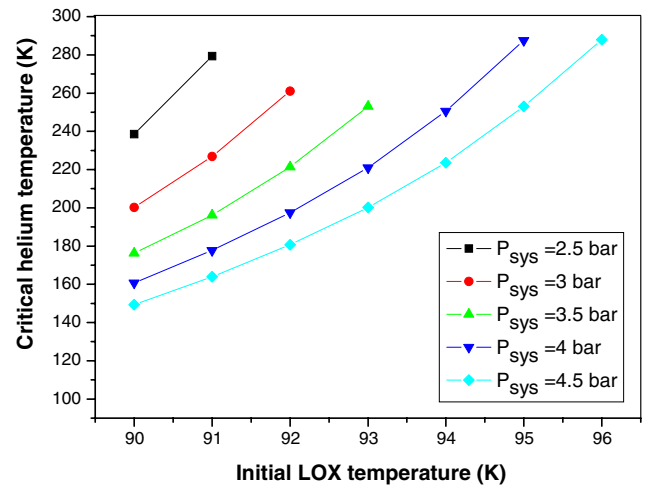


Fig. 17. Critical helium temperature with initial LOX temperature variation under pressurized conditions.

oxygen, with liquid oxygen initial temperature variation using Eq. (30). As initial liquid oxygen temperature increases and system pressure decreases, X_{cr} value decreases. X , which is the combination of relevant operating parameters, should be established as higher value than X_{cr} for cooling liquid oxygen.

Fig. 17 shows the calculation results for critical helium temperature, defined as the possible maximum injected helium temperature for cooling liquid oxygen, with liquid oxygen initial temperature variation using Eq. (31). It is assumed that helium is injected with the specified gas flow rate ($\dot{V}_{He}/V_{LOX} = 1 \text{ min}^{-1}$). As the system pressure increases, the critical injected helium temperature is lowered and as the initial liquid oxygen temperature increases, the critical injected helium temperature increases. For initial liquid oxygen temperatures higher than those in the range as shown in Fig. 17, room-temperature helium gas without precooling is just sufficient for cooling liquid oxygen. Injected helium temperature should be set to be lower than the critical helium temperature to cause cooling effect.

7. Summary

Characteristics of helium injection cooling into liquid oxygen under pressurized condition are thoroughly examined by theoretical, analytical and experimental methods. The ‘finite heat transfer and instantaneous diffusion mass transfer’ model is applied for the analysis. The typical characteristic of helium injection under pressurized condition compared to atmospheric condition is that the liquid oxygen is usually sub-cooled and relatively higher helium mass flow is supplied with specific volume flow rate. For initially saturated liquid oxygen, cooling effect is achieved regardless of the injected helium temperature and the flow rate even under pressurization condition. For initially sub-cooled liquid oxygen, the developed gaseous oxygen

partial pressure may exceed the corresponding liquid oxygen saturation pressure which leads to the failure of cooling. Lowering injected helium temperature is an effective means for lowering gaseous oxygen partial pressure in helium bubble.

The gaseous oxygen partial pressure developed in initial injection stage is the main parameter for determining cooling condition. A simplified model to estimate gaseous oxygen partial pressure and subsequent cooling rate at initial helium injection stage is developed. The model assumes that the gaseous oxygen partial pressure is initially developed only by heat transfer between helium gas and liquid oxygen and then liquid oxygen vaporization due to diffusion occurs due to the difference of gaseous oxygen partial pressure. A non-dimensional parameter representing the partial pressure ratio is suggested and applied for several cases to identify the cooling regime. Critical value of the non-dimensional parameters and injected helium temperatures are also investigated. The proposed model successfully reflects cooling or no cooling results obtained from the experimental data and it is considered as a useful guide for determining the cooling condition during preliminary design stage of effective helium injection system.

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